Slide 1 — Pipeline: How the Tool Works

- Input: per-cell response scores + metadata from Tahoe-100M H5AD (plate, well, cell line, drug, dose, DMSO role), plus precomputed pseudobulk summaries in Parquet for the interactive site.
- User selects one cell line × drug; distribution of response signal is plotted per dose, referenced against plate-matched DMSO
- Two paired metrics per condition:
  - *Magnitude* — how far the median shifted from DMSO (the statistic is the median difference standardized by the DMSO median absolute deviation.)
  - *Coverage* — fraction of treated cells outside the DMSO-calibrated reference interval; residual fraction is the fraction remaining inside it.
- Dispersion check — does treatment shift, widen, compress, or split the distribution, not just move its center
- Confounder check — (QC sensitivity) test whether the apparent residual fraction is associated with RNA depth, detected genes, mitochondrial fraction, cell-cycle state or sample identity.
- Pairwise dose-vs-DMSO comparison; The response-completeness pipeline has not yet been run on the real Tahoe single-cell data. The 65,218-condition public Atlas is a separate real pseudobulk analysis.

Slide 2 — Statistical Design

- Two-plate replicate design: Plate 6 = development, Plate 14 is examined only after the rule is frozen.
- On Plate 6, independent DMSO samples are assigned calibration and held-out negative-control roles where the metadata confirms the design, never pooled: one → calibration reference, one → held-out negative control.
- Response score is not yet frozen. A pathway or mechanism-of-action score is the preferred starting point, but the exact score, reference interval alpha and QC rules remain to be selected using plate 6 only.
- DMSO-vs-DMSO and treatment-vs-DMSO comparisons run in paraables, and every threshold logged.
- Full rule frozen — score definition, DMSO roles, reference interval, QC cutoffs — before Plate 14 is opened; Plate 14 must not be inspected for parameter selection if we retain it as validation.
- *Two caveats explicitly reported:*
  - The Wilson intervals describe within-sample uncertainty in a cell fraction. Biological replication comes from the plate 6/14 comparison.
  - Current replicate matches cover (mainly) the high-dose condition, not a full dose series — dose-consistency stated as a separate claim.

Slide 3 — Implementation & Reproducibility

- The public site now has two distinct parts: a real pseudobulk Atlas covering 65,218 conditions, and an illustrative single-cell response-completeness interface.
- The real Atlas is valid, but it does not yet estimate residual fractions or cell-level heterogeneity.
- Results export to a shared, versioned JSON schema — the interface currently reads from this.
- Public site (GitHub Pages) already serves a real pseudobulk Atlas (65k+ conditions: ranking, compare, signature search); this schema is what lets the first real response-completeness result replace the illuesigning anything

- Current status: response-completeness specification and software infrastructure tested and merged; exact score and calibration configuration pending metadata inspection; first real plate-6 run not yet completed.